Rigid Analytic Geometry Week 8 Solution

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Problem 1 By Problem 8 of the previous sheet, there is $R \in (r, 1]$ such that for $r < |x| \leq R$, we have,

$$|f_i(x)| = n_i \left(\frac{|x|}{r}\right)^{j_i}.$$

Since the ordering is lexicographic, we have two cases for each $i = 1, \cdots, m$, such that

1. $n_i < n_0$
2. $n_i = n_0$ and $j_i \leq j_0$.

For the first case,

$$n_i \left(\frac{R}{r}\right)^{j_i} < n_0 \left(\frac{R}{r}\right)^{j_0}.$$

holds by taking suitably small R and using the continuity of $|\cdot|$, we obtain the inequality.

For the second case, $j_i \leq j_0$ implies

$$\left(\frac{|x|}{r}\right)^{j_i} < \left(\frac{|x|}{r}\right)^{j_0}.$$

This shows that

$$r < |x| \leq R \Rightarrow x \in \mathcal{R}(f_0|f_1, \cdots, f_m).$$

This proves the first part.

For the second part, we again have two cases,

1. $n_i > n_0$
2. $n_i = n_0, j_i > j_0$.

For the first case, again by the continuity of $|\cdot|$, we have,

$$n_i \left(\frac{R}{r}\right)^{j_i} > n_0 \left(\frac{R}{r}\right)^{j_0}.$$

And similarly for the other case. Therefore,

$$|f_i(x)| > |f_0(x)|$$

for all $r < |x| \leq R$. Therefore,

$$\Omega \cap B_{\leq R}(0) \subseteq B_{\leq r}(0).$$

Problem 5
By Problem 4, we can take $F = \{x_1, \cdots, x_r\} \subset \mathcal{S}(B)$ such that

$$z \in \bigcap_{j=1}^{r} S_R(x_j) \ni Z \Rightarrow |f_0(z)| = \|f_0\|_B.$$

Then by the definition of norms and the assumption,

$$|f_i(x)| \leq \|f_i\|_B \leq \|f_0\|_B = |f_0(0)|, 1 \leq i \leq m.$$

This shows that

$$\bigcap_{S \in F} S \subseteq \Omega.$$

The second part is similarly shown by applying Problem 4 to f_i such that

$$\|f_0\|_B < \|f_i\|_B.$$

That is

$$x \in \bigcap_{S \in F} S \Rightarrow |f_0(x)| < |f_i(x)|.$$

Therefore,

$$\Omega \cap \bigcap_{S \in F} S = \emptyset.$$

Problem 7
Set,

$$\Phi : A \times M \to B \times M, (a, m) \mapsto (a \cdot 1_B, m).$$

Since $a \mapsto f(a)$ is continuous Φ is continuous. By assumption $\rho : B \times M \to M, (b, m) \mapsto bm$ is continuous. Therefore

$$\rho \circ \Phi : A \times M \to M$$

is continuous. This shows M is an adic A-module.

Tutor's solutions

Problem 1,2 Accorind to Sheet 9 Problem 8, one can replace there is $r_+, r_-, 0 < r_- < r < r_+$ in Sheet 7 Problem 1,3 by there is $R, r < R < 1$ and the proof is the same.

Problem 3 We have,

$$B = \bigcap_{S \in \mathcal{S}} S \neq \emptyset,$$

where $B \in \xi$. if S, S' are different elements of $B = K^\circ$, Take $S \neq S' \in \mathcal{S}(B)$ Say $S = S_1(x), S' = S_1(y)$. Then $|x-y| = 1$, else $S = S'$. For any $z \in B, |z-x| < 1$ and $|z-y| < 1$, cannot happen at the same time. Therefore $z \in S$ or $z \in S'$. Thus $S \cup S' = B$. $B \in \xi \Rightarrow$ one of S, S' is in ξ.

Remark 0.1. *Any finite intersection of elements in $\mathcal{S}(B)$ is non-empty.*

Proof. Assume $B = K^\circ$. Take$x \in K^\circ$ Then $|y-x| = 1$ if and only if $y-x \notin K^{\circ\circ}$. Take $x_1, \cdots, x_m \in K^\circ$. Then,

$$\bigcap_i \mathcal{S}_1(x_i) = \{y \in K^\circ \mid \overline{y} \neq \overline{x}_1, \cdots, \overline{x}_m \text{ in } k = K^\circ/K^{\circ\circ}\}.$$

Note that K is algebraically closed, so $|k| = \infty$. Therefore $\bigcap_i S_1(x_i) \neq \emptyset$. $\square$

Problem 4 Assume $\|g|\mathbb{T}_1\| = 1$. Then $|g(x)| < \|g|\mathbb{T}_1\|$ if and only if $g(x) \in K^{\circ\circ}$ if andon ly if $\overline{x}$ is a root of $\overline{g}$ in $k = K^\circ/K^{\circ\circ}$. Note that $\overline{g}$ is a polynomial in k, so we may take all of its roots $\alpha_1, \cdots, \alpha_n$. Choose x_i to be a lift of α_i, then

$$|g(x)| < \|g|\mathbb{T}_1\| \iff x - x_i \in K^{\circ\circ} \iff x \in \bigcup_i (x_i + K^{\circ\circ}).$$

Take the complement.

Problem 5 Take a finite set $RF \subseteq \mathcal{S}(B)$ such that for $x \in \bigcap_{S \in F} S$,

$$|f_i(x)| = \|f_i|\mathbb{T}_1\|.$$

Then the conclusion is immediate.

Problem 6 For a rational open Ω, $\Omega \in \xi_B \Rightarrow \exists F \subseteq \mathcal{S}(B)$ finite with

$$\bigcap_{S \in F} F \subseteq \Omega.$$

Since each S is in ξ, so does their finite intersection. Hence $\Omega \in \xi$.

$\Omega \notin \xi_B$, then there is a finite set $F \subseteq \mathcal{S}(B)$ such tat

$$\Omega \cap \bigcap_{S \in F} S = \emptyset.$$

Note that $\bigcap_{S \in F} S \in \xi$ therefore $\Omega \notin \xi$. Thus $\xi = \xi_B$.

Problem 7

Lemma 0.1. *If A is a Huber ring, $(A^\#, I)$ be a pair of definition, C be an open subring of A, then*

$$(C \cap A^\#, I^N A^\#)$$

is also a pair of definition.

Proof. Since $C \cap A^\#$ is open, there is N such that $I^N A^\# \subseteq C \cap A^\#$. Then $I^N A^\#$ is finitely generated $C \cap A^\#$-ideal. First

$$(I^N A^\#)^n (C \cap A^\#)$$

is open. $\square$

Moreover, $I^N A^{\#}$ is finitely generated, also $(I^N A^{\#})^n (C \cap A^{\#})$ gives 0-neighbourhood of A.

Problem 7

Lemma 0.2. *If M is an adic A-module with respect to $(A^{\#}, I)$. Let (A', J) be another pair of definition. Then M is adic with respect to (A', J).*